Dealer Intermediation Capacity and U.S. Treasury Auction Pricing

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Research Question, Approach, and Data

My research question is: **How do primary dealer intermediation conditions affect the pricing of U.S. Treasury auctions?** Primary dealers have traditionally played an important role in absorbing new Treasury supply and providing liquidity in the Treasury market. If dealers already hold large inventories, their ability or willingness to absorb additional securities may become weaker, and an auction may require a larger price concession. This question is especially relevant today because Treasury issuance has increased while the composition of auction demand has also changed.

My basic empirical strategy is to match each Treasury auction with the most recent primary dealer inventory in the corresponding maturity bucket and test whether auction outcomes become weaker when dealer inventories are relatively high. My preferred final pricing measure is the conventional auction tail, defined as the auction stop yield minus the when-issued yield immediately before the auction. I currently have Bloomberg Terminal access, but historical Excel downloads for when-issued data are still returning "N/A Review." Instead of waiting for this issue to be solved, I have already implemented the research design using three outcomes that can be constructed entirely from the auction itself: the high-minus-median accepted yield spread, the bid-to-cover ratio, and the primary dealer allotment share. I also use constant-maturity Treasury yields for a separate event-study analysis around auction dates.

I have already collected and merged the main public data. Treasury auction results come from TreasuryDirect and the Treasury Fiscal Data API. The raw sample contains 1,213 Note and Bond auctions from January 2015 through October 7, 2026. After removing 142 TIPS auctions, 143 floating-rate note auctions, one incomplete auction, and two very small special reopenings below $1 billion, the final sample contains **925 completed nominal fixed-rate Treasury auctions**. Primary dealer positions come from the Federal Reserve Bank of New York Primary Dealer Statistics and are matched to auctions by maturity. Daily Treasury yields come from FRED. My sample spans three New York Fed series segments beginning in 2015, 2022, and 2024. The main classification change that I can directly observe occurs in 2022, when the long-maturity category above 11 years is separated into 11–21 years and above 21 years. I account for these series segments when constructing the weekly inventory measures.

Literature Review and Empirical Motivation

Lou, Yan, and Zhang (2013) document a Treasury auction cycle in which Treasury prices decline before auctions and recover afterward, even though auction timing and size are known in advance. They connect this pattern to limited intermediary risk-bearing capacity and slow-moving investor capital. Their findings provide the basic motivation for asking whether predictable Treasury supply

can still create temporary price pressure in a highly liquid market.

Fleming, Nguyen, and Rosenberg (2024) study how Treasury dealers manage their inventories using dealer position data from 1990 to 2020. They find that Treasury issuance is an important driver of weekly dealer inventory changes. Dealers do not immediately remove all auction-related inventory and can receive compensation for carrying this inventory risk through later price appreciation. They also find that dealers reduce position taking and lay off inventory faster when balance-sheet costs are higher. Their results motivate my use of dealer inventory as a measure of intermediation conditions.

More recent research measures dealer constraints more directly. Bräuning and Stein (2024) use confidential dealer-level data and study shocks to supplementary leverage ratio and value-at-risk constraints. They find that tighter constraints reduce dealer Treasury positions and market turnover, increase intermediation margins, and weaken Treasury auction outcomes. This provides strong evidence that dealer balance-sheet capacity can matter for Treasury-market pricing, but their strongest identification depends on confidential microdata that are not available in public datasets.

Fleming, Liu, and Nguyen (2026) provide high-frequency evidence on Treasury auction price pressure. Using intraday Treasury data, they find that yields rise before auctions and reverse afterward. The effect is stronger when dealer risk-bearing capacity is tighter and weaker when investor demand is stronger. They also find that growing non-dealer participation has helped the market absorb greater Treasury issuance.

This change in auction participation is very visible in my own sample. The average primary dealer allotment share falls from about **34.9 percent in 2015 to 10.2 percent in 2026**, while the direct bidder share rises from 11.5 percent to 20.8 percent. At the same time, the bid-to-cover ratio does not show a similar long-run decline. Treasury supply continues to be absorbed, but the investors absorbing it have changed.

This creates an important question for the recent Treasury market. Existing research gives strong evidence that dealer constraints can matter for pricing, but much of the dealer-centered mechanism was studied during periods when dealers absorbed a larger share of auction supply. My sample asks whether dealer intermediation conditions remain visible in auction pricing when dealers are no longer the same marginal direct buyers they were earlier.

There are also two measurement problems. First, high aggregate dealer inventory does not necessarily mean that a balance-sheet constraint is binding. Dealers may hold large positions voluntarily because they expect profitable price movements or because of customer activity. Second, broad constant-maturity Treasury yield changes are not the same as the concession on the specific security being auctioned. These two problems motivate my empirical design.

Contribution and Measurement Improvement

The first improvement in my approach is to measure dealer inventory by maturity rather than use

one total Treasury position. I match 2-year auctions with the dealer position bucket of 2 years or less, 3-year auctions with 2–3 years, 5-year auctions with 3–6 years, 7-year auctions with 6–7 years, and 10-year auctions with 7–11 years. At the long end, the reporting structure changes in 2022. Before 2022, positions above 11 years are reported together. After 2022, they are separated into 11–21 years and above 21 years. I use these categories to construct the closest possible inventory match for 20-year and 30-year auctions.

The second improvement is the measurement of auction pricing. My preliminary event study uses constant-maturity Treasury yields. These yields can show whether price pressure exists around auctions, but they also include CPI releases, employment reports, FOMC information, oil-price changes, and other market news. They also do not represent the exact CUSIP being auctioned. My preferred final outcome is therefore:

$$\text{Auction Tail} = \text{Auction Stop Yield} - \text{Pre-auction When-issued Yield}$$

The when-issued yield measures the market yield of the specific security immediately before the auction. Bloomberg Terminal access has been approved, but the historical Excel download is currently returning "N/A Review." In the meantime, I have advanced the analysis using auction-internal outcomes that are calculated directly from auction results rather than from multi-day market yield movements.

The third issue is that inventory and financial constraints are not the same variable. I investigated possible dealer financing measures. The New York Fed reports weekly dealer repo and reverse-repo financing amounts, but many observations after 2022 are suppressed for confidentiality. The publicly available GCF Repo Index also has only a short recent history, while a longer history would require Bloomberg. Therefore, dealer financing conditions will be treated as an extension rather than part of my baseline specification. The quarterly Senior Credit Officer Opinion Survey on Dealer Financing Terms may provide a lower-frequency robustness measure.

This paper studies whether dealer intermediation conditions remain visible in Treasury auction pricing during a recent period when primary dealers absorb a much smaller share of new issuance. I use transparent public data, maturity-matched dealer inventories, and multiple auction-level measures of pricing and demand. The project therefore asks both whether the traditional dealer-capacity mechanism remains economically relevant in the changing Treasury auction market and how much of this mechanism can be observed using aggregate public data.

Data and Sample Construction

My main unit of observation is an individual Treasury auction. The Treasury Fiscal Data API initially provides 1,213 Note and Bond auctions beginning in January 2015. Because the raw data include different types of Treasury securities, sample construction is important.

I remove 142 auctions for which inflation_index_security = Yes, because these are TIPS and their auction yields are real rather than nominal yields. I remove another 143 observations for which floating_rate = Yes. There are no callable securities or cash-management bills in this sample. I also remove one auction with missing auction results because it had not yet taken place when the data were downloaded. Finally, I remove two special reopenings below $1 billion because their sizes are extremely different from regular Treasury auctions. The locked sample contains **925 nominal fixed-rate coupon auctions from January 12, 2015 through October 7, 2026**.

The final sample contains 131 2-year auctions, 141 3-year auctions, 142 5-year auctions, 150 7-year auctions, 143 10-year auctions, 77 20-year auctions, and 141 30-year auctions. The smaller 20-year sample reflects the reintroduction of the 20-year Treasury in 2020.

The auction dataset provides several outcomes that I can use without Bloomberg. The first is the high-minus-median accepted yield spread. Across all 925 auctions, this variable has a mean of **5.253 basis points**, a standard deviation of 1.664 basis points, and a range from 1.50 to 11.90 basis points. I do not call this variable an auction tail because it does not compare the auction result with the pre-auction market price.

The second outcome is the bid-to-cover ratio. Its sample mean is 2.523 and its standard deviation is 0.213. The third group of variables measures bidder allocation. The average primary dealer allotment share is 20.6 percent, the average indirect bidder share is 55.9 percent, and the average direct bidder share is 14.8 percent. I calculate these allotment shares using the total accepted amount as the denominator, consistent with the shares displayed in my descriptive figures.

The second major dataset is the New York Fed Primary Dealer Statistics. I use weekly outright positions in Treasury coupon securities excluding TIPS. The data span New York Fed reporting series beginning in 2015, 2022, and 2024. The main directly observable classification change is the 2022 split of the greater-than-11-year maturity bucket into 11–21 years and above 21 years. For the long end, I combine the two newer categories where necessary to construct a consistent measure across time.

For each auction, I assign the most recent dealer inventory observation available before the auction. Inventory is highly persistent and has a strong long-run trend, so I standardize it within **maturity** $\times$ **year cells**. This measures whether inventory is unusually high or low relative to other observations for the same maturity in the same year.

Auction size is measured using offering_amt, the public offering amount announced before the auction. I also construct a relative size variable equal to the current offering amount divided by the average offering amount for the same maturity and year. This is different from total_accepted, which I use as the denominator for bidder allotment shares.

Finally, I use daily Treasury constant-maturity yields from FRED to construct auction event windows. The trading-day index is based on non-missing yield observations so that federal holidays are not incorrectly counted as trading days.

Descriptive Evidence and Auction Cycle

The first important descriptive result is the change in bidder composition. In 2015, primary dealers received approximately 34.9 percent of auction supply, compared with only 10.2 percent in 2026. Direct bidders increased from approximately 11.5 percent to 20.8 percent, while indirect bidders also became more important. At the same time, the bid-to-cover ratio remained relatively stable. This suggests that Treasury supply continues to be absorbed, but the composition of investors has changed substantially (Figure 1).

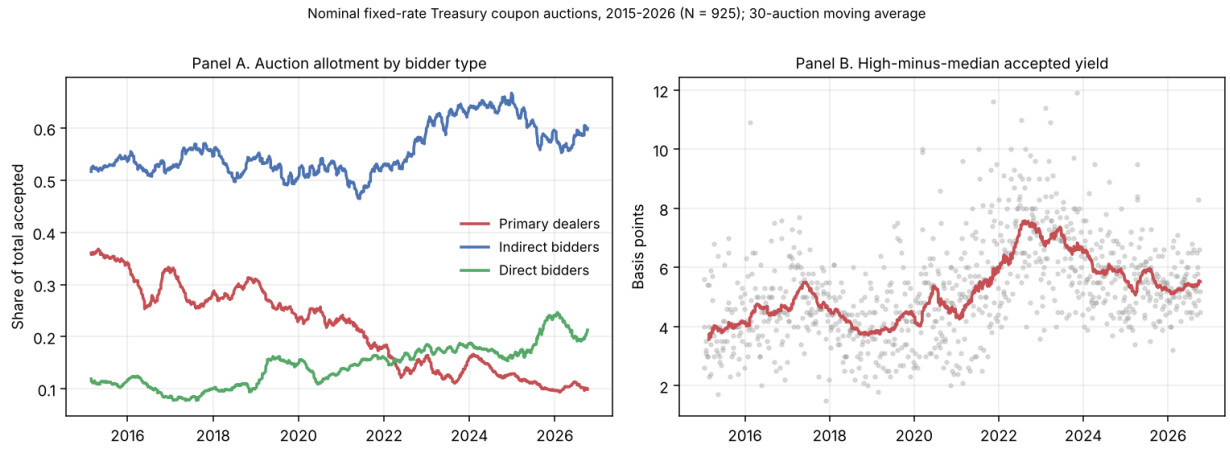


Figure 1. Auction allotment by bidder type and the high-minus-median accepted yield spread. The sample includes 925 nominal fixed-rate Treasury coupon auctions from 2015 to 2026. Lines show 30-auction moving averages.

The second descriptive result is the Treasury auction cycle. I examine three event windows using daily constant-maturity Treasury yields: the five trading days ending on the auction date, the five trading days ending one day before the auction, and the ten trading days ending on the auction date.

For the 10-year Treasury, the average yield increase from t−5 to t0 is **2.66 basis points**, with a t-statistic of **2.91**. The alternative t−6 to t−1 window gives an increase of 2.38 basis points (t = 2.53). For the 30-year Treasury, the corresponding estimates are **2.99 basis points (t = 3.30)** and **2.15 basis points (t = 2.31)**. Both maturities show statistically significant yield increases under the two five-day event windows (Figure 2).

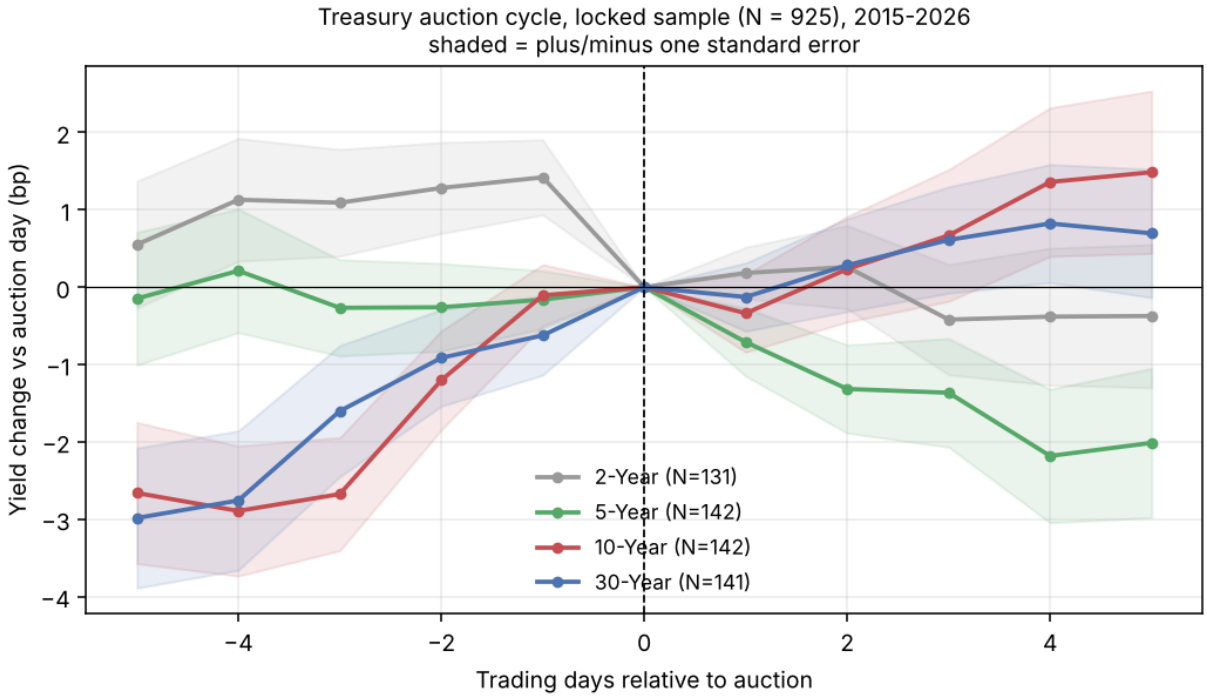


Figure 2. Treasury yield movements around auctions for selected maturities (2Y, 5Y, 10Y, and 30Y). Yield changes are measured relative to the auction-day yield, in basis points. Shaded bands represent plus or minus one standard error.

The pattern across maturities is less stable. The 2-year yield increases significantly in the t−6 to t−1 window but changes sign in the t−5 to t0 window. The 20-year Treasury shows stronger evidence in the ten-day window. I therefore interpret these findings conservatively: there is evidence of pre-auction yield pressure, especially for 10-year and 30-year securities, but the exact pattern depends on the event-window definition.

These event-study results do not directly answer my research question because constant-maturity yields also reflect macroeconomic news unrelated to Treasury auctions. I therefore use auction-internal pricing and bidding outcomes in the main regressions.

Main Regression Design and Results

My main explanatory variable is standardized, maturity-matched dealer inventory. The current baseline regression is:

$$Y_i = \alpha + \beta\,\text{InventoryZ}_{m,t-1} + \gamma\,\text{RelativeSize}_i + \mu_m + \lambda_y + \varepsilon_i$$

Here, Y_i is alternatively the high-minus-median accepted yield spread, the bid-to-cover ratio, or the primary dealer allotment share. InventoryZ is the most recent maturity-matched dealer inventory before the auction, standardized within maturity and year. RelativeSize is the announced offering amount relative to the average offering amount for the same maturity and year. The model also includes maturity fixed effects and year fixed effects.

The coefficient of main interest is β. If high dealer inventory reflects tighter intermediation capacity, I would expect higher inventory to be associated with weaker auction pricing, such as a larger high-minus-median spread or a lower bid-to-cover ratio.

The expected sign for the dealer allotment share is less clear because two mechanisms can operate in opposite directions. Under a balance-sheet constraint mechanism, high inventories may cause dealers to bid more conservatively and receive a smaller share of the auction. Under a residual absorption mechanism, weak final-investor demand may force dealers to absorb more securities, causing the dealer share to increase. Because these mechanisms imply opposite signs, I treat dealer allotment share as a supporting allocation outcome rather than as the main test of the dealer-capacity hypothesis.

I estimate four specifications: inventory only; inventory plus relative auction size; the same model with maturity fixed effects; and finally maturity and year fixed effects. In the full specification, the inventory coefficient for the high-minus-median spread is **−0.0349 basis points with a t-statistic of −0.82**. For the bid-to-cover ratio, the coefficient is **−0.0073 with a t-statistic of −1.37**. For the dealer allotment share, the coefficient is **+0.0024 with a t-statistic of +1.36**. None of these estimates is statistically significant.

The null result also appears across maturities. I estimate separate regressions for the 2-, 3-, 5-, 7-, 10-, and 30-year sectors. Across 18 maturity-specific coefficients for the three outcomes, only two have absolute t-statistics above 2, and the significant coefficients do not consistently have the sign predicted by the original inventory hypothesis.

I also split the sample into 2015–2019, when dealers still received about 29 percent of auction supply on average, and 2020–2026, when the average share is about 15 percent. For the high-minus-median spread, the inventory coefficient is −0.0180 with a t-statistic of −0.30 in 2015–2019 and −0.0576 with a t-statistic of −1.01 in 2020–2026. The interaction between inventory and the early-period indicator is also insignificant.

This suggests that the null result is not simply caused by the recent decline in dealer auction participation. Even during the earlier period, when dealers still absorbed close to one-third of supply, aggregate dealer inventory does not predict auction pricing in my data.

The current regressions use standard errors clustered by auction date. Because dealer inventory is observed weekly, I will also use week-level clustering as a robustness test because multiple auctions can share the same weekly dealer-state observation.

Demand Composition, Interpretation, and Next Steps

Although dealer inventory has little relationship with auction outcomes, bidder composition is strongly associated with auction pricing. I standardize indirect and direct bidder shares within maturity and year and include them in the same regression with dealer inventory, relative auction size, maturity fixed effects, and year fixed effects.

For the high-minus-median spread, a one-standard-deviation increase in the indirect bidder share is associated with a **0.280 basis point lower spread**, with a t-statistic of −6.58. A one-standard-deviation increase in direct bidder participation is associated with a **0.180 basis point lower spread**, with a t-statistic of −4.01. For the bid-to-cover ratio, the indirect bidder coefficient is +0.0410 with a t-statistic of 7.18, while the direct bidder coefficient is +0.0139 with a t-statistic of 2.45. Dealer inventory remains weak in the same regressions.

These estimates are descriptive correlations, not causal effects. Bidder shares and auction outcomes are determined in the same auction. When demand is strong, indirect investors may receive more securities while the auction also has a smaller high-minus-median spread and a higher bid-to-cover ratio. Therefore, I do not interpret the result as evidence that higher indirect bidder participation causes stronger auction pricing. Instead, the result shows that demand composition moves much more closely with auction pricing than the aggregate dealer inventory measure.

There are at least three possible interpretations of the dealer-inventory null result. First, aggregate dealer inventory may be a poor proxy for whether balance-sheet constraints are actually binding. Confidential dealer-level measures contain information that weekly aggregate positions cannot provide. Second, although the auction-internal outcomes are more directly related to auction bidding than multi-day Treasury yield changes, they are still not the conventional auction tail. Historical when-issued yields could provide a more precise pricing measure. Third, dealer inventory may genuinely have become less important for marginal auction pricing as a larger share of Treasury supply is absorbed directly by non-dealer investors.

The Bloomberg when-issued data will help distinguish the second explanation. Terminal access is available, but the historical Excel download is currently returning "N/A Review." Once the series is available, my preferred specification will be:

$$\text{AuctionTail}_i = \alpha + \beta\,\text{InventoryZ}_{m,t-1} + \gamma\,\text{RelativeSize}_i + \mu_m + \lambda_y + \varepsilon_i$$

If β is positive, high dealer inventories are associated with larger auction concessions relative to the pre-auction market price. If the coefficient remains close to zero even with a security-specific when-issued tail, the conclusion will be stronger: publicly observable aggregate dealer inventory may not be a useful measure of the dealer-level intermediation constraints that affect modern Treasury auction pricing.

Dealer financing conditions remain another possible extension, but public data have important limitations. The New York Fed's weekly repo and reverse-repo financing series contain substantial

confidentiality-related suppression after 2022, while the public GCF history is too short for the full sample. I therefore do not include financing conditions in the baseline specification. The quarterly SCOOS survey may provide a lower-frequency robustness check.

At this stage, the evidence has changed how I view my original hypothesis. I began with the idea that high dealer inventory should predict weaker Treasury auctions. The auction-cycle evidence shows that yields often increase around Treasury issuance, but aggregate dealer inventory does not explain the auction-internal pricing outcomes that I can currently observe. At the same time, the composition of auction demand has changed substantially, and bidder participation is strongly related to pricing outcomes. The next stage of the project will test whether the inventory null remains after measuring the auction concession with when-issued yields and will examine more carefully whether the changing role of non-dealer investors helps explain recent Treasury auction pricing.

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